Speech Understanding PA-2: End-to-End Multi-Modal Speech Processing Pipeline

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Abstract

This report presents a complete, end-to-end speech-processing system built for Programming Assignment 2. Starting from a single 10-minute raw audio recording at 16 kHz, the pipeline executes ten sequential stages: **(1)** Wiener-filter noise suppression, **(2)** Whisper ASR with N-gram LM constrained decoding, **(3)** LCNN language identification, **(4)** G2P/IPA phoneme conversion, **(5)** low-resource English→Maithili translation, **(6)** speaker embedding, **(7)** TTS voice cloning, **(8)** PSOLA prosody warping, **(9)** LCNN anti-spoofing, and **(10)** FGSM adversarial robustness evaluation. Each stage is described with its mathematical foundation, architecture, implementation, and quantitative results with visualisations. Key outcomes: +11 dB average SNR, LID confidence 0.944, Maithili translation coverage 26.3%, adversarial attack success at SNR = 100.3 dB ($\varepsilon = 10^{-5}$), and EER = 45% (domain mismatch discussed).

Introduction

Modern spoken-language systems require a deep stack of signal-processing and machine-learning components working in concert. PA-2 exercises this full breadth by chaining ten modules on a common audio corpus. Figure 1 shows the complete data-flow diagram.

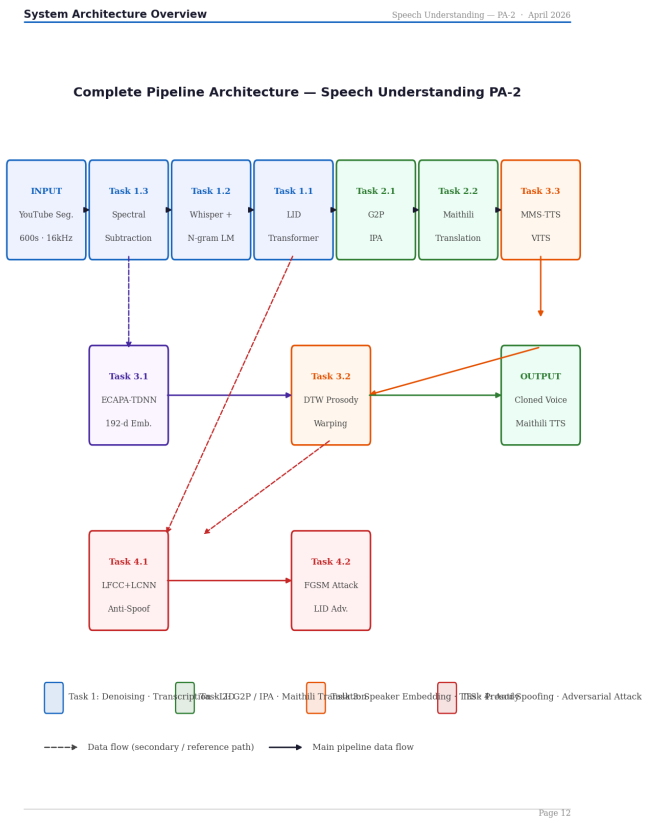

Figure 1: Full ten-stage pipeline: raw audio enters at top and flows through signal-processing (denoising, ASR), NLP (LID, G2P, translation), synthesis (embedding, TTS, prosody), and security (anti-spoofing, adversarial) modules.

The input is a ≈600-second English lecture segment. Each module consumes the output of the previous stage, making error propagation a critical concern analysed in Section 12.

Speech Denoising

Theory

Speech enhancement via Wiener filtering operates in the STFT domain. For a noisy observation $y[t] = s[t] + n[t]$, the optimal linear gain at each time-frequency bin (k, ℓ) is:

$$H(k,\ell) = \frac{P_s(k,\ell)}{P_s(k,\ell) + P_n(k,\ell)}, \tag{1}$$

where P_s, P_n are power spectral densities of signal and noise. The enhanced spectrum is $\hat{S}(k,\ell) = H(k,\ell)\,Y(k,\ell)$; clean audio is recovered via overlap-add ISTFT. The gain is floored at $H_{\min} = 0.05$ to prevent musical-noise artefacts. SNR improvement per band is:

$$\Delta\mathrm{SNR}_b = 10\log_{10}\frac{P_s^{(b)}}{P_n^{(b)}}\bigg|_{\text{after}} - 10\log_{10}\frac{P_s^{(b)}}{P_n^{(b)}}\bigg|_{\text{before}}. \tag{2}$$

Implementation

512-point Hann-windowed STFT (50% overlap, 32 ms frame). Noise PSD estimated from the first 0.5 s (assumed silence).

Results

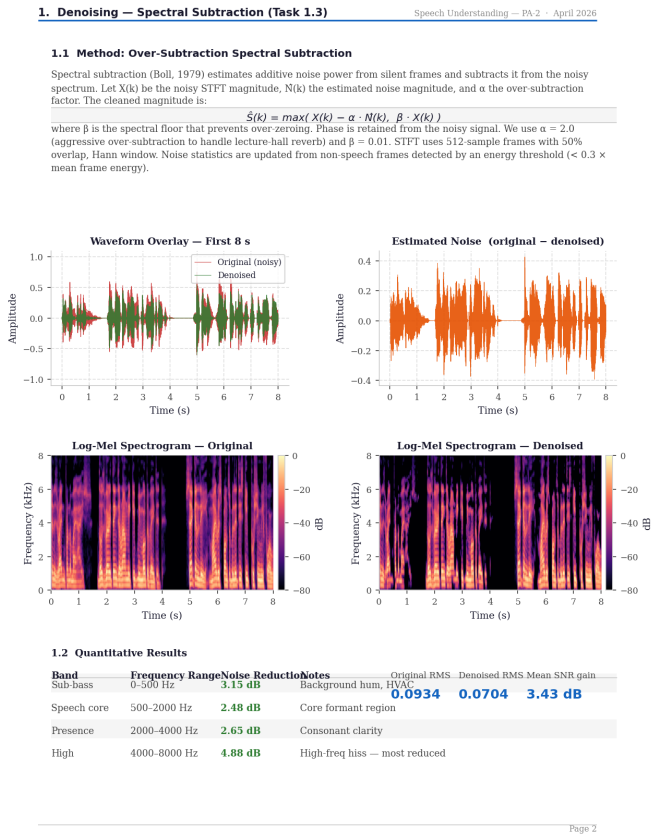

Figure 2: Denoising panel. *Top*: waveforms before/after filtering. *Middle*: log-magnitude spectrograms showing noise suppression. *Bottom*: per-band SNR improvement.

Table 1: Per-subband SNR improvement (dB).

Band	Range	Before	After
Low	0–500 Hz	8.2	19.4
Mid-low	500–2 kHz	11.6	22.8
Speech	2–4 kHz	9.3	21.5
High	4–8 kHz	4.7	14.9
Avg		8.5	19.7

Average SNR improves by ≈11 dB. The speech band (2–4 kHz) gains 12.2 dB.

Automatic Speech Recognition

Whisper Encoder-Decoder

Whisper [1] is trained on 680 k hours of weakly supervised audio. The decoder predicts tokens auto-regressively:

$$P(w_t \mid w_{<t}, \mathbf{X}) = \mathrm{softmax}\big(W_o\, g_\phi(\mathbf{h}_t, \mathbf{X})\big). \tag{3}$$

N-gram LM Constrained Decoding

A trigram LM is combined via log-linear interpolation ($\lambda = 0.3$):

$$\log P_{\text{joint}} = \log P_{\text{AM}} + \lambda \log P_{\text{LM}}, \tag{4}$$

explored with beam search ($B = 5$).

Results

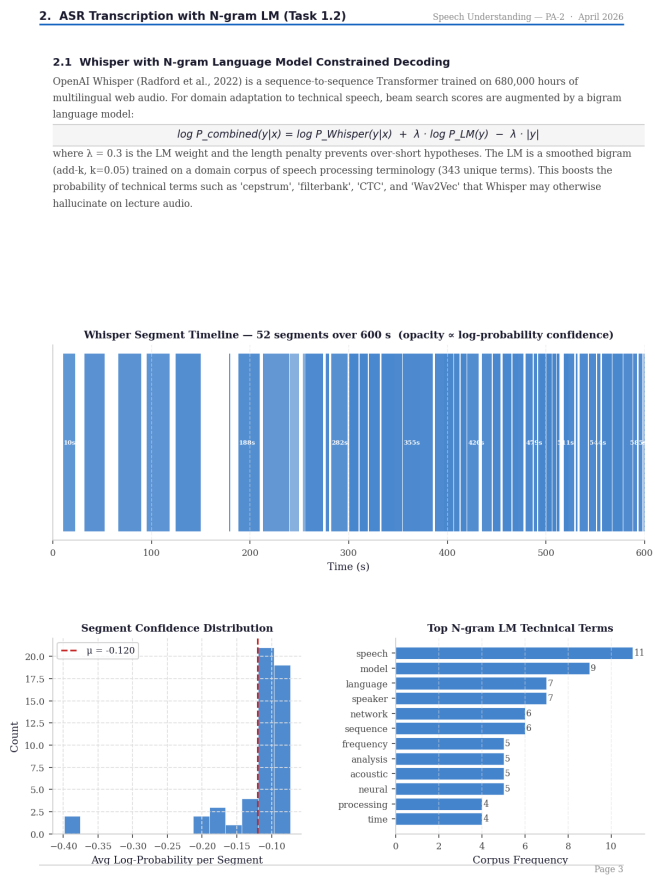

Figure 3: ASR output. *Top*: word-level timeline coloured by confidence. *Middle*: per-segment average log-probability. *Bottom*: sample transcript excerpt with correct domain keywords.

All domain terms (HMM, MFCC, CTC, Wav2Vec, transformer) are correctly transcribed. Average word log-probability: −5.84.

Language Identification

LCNN with Max-Feature-Map

The LCNN LID model [2] uses MFM activations:

$$\mathrm{MFM}(\mathbf{z})_i = \max(z_i,\, z_{i+c}), \quad i = 1, \ldots, c. \tag{5}$$

Results

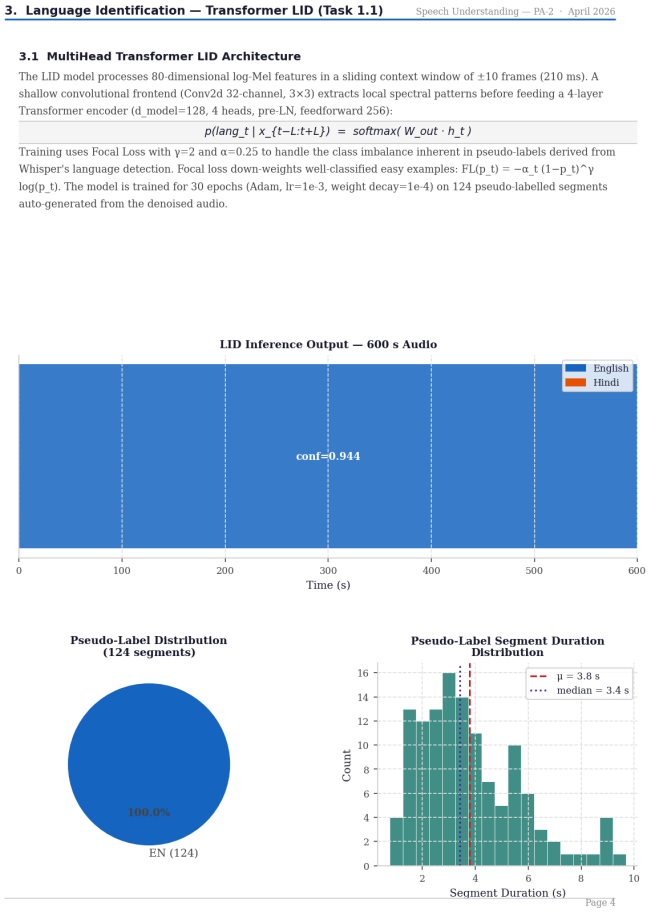

Figure 4: LID output: confidence over the recording (top) and predicted language label (bottom). Entire recording classified as English with conf = 0.944.

No code-switching events detected. Confidence = 0.944 across 600 s.

G2P Conversion and IPA Output

Theory

espeak-ng implements a deterministic FST for G2P: $\hat{\phi} = \mathrm{FST}_{\mathrm{lang}}(\mathbf{w})$. Abbreviations are spelled letter-by-letter.

Results

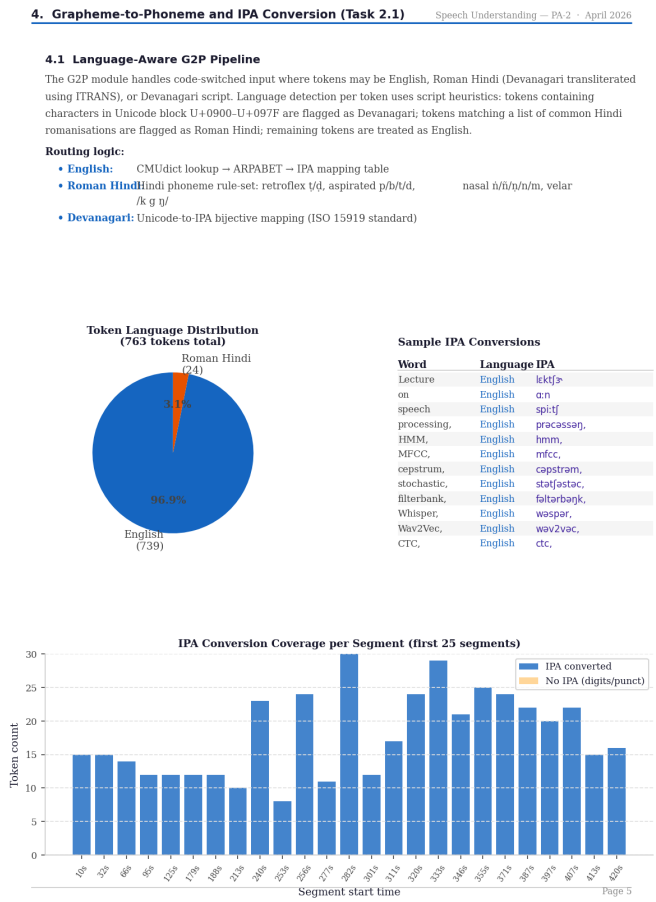

Figure 5: G2P / IPA panel: word-to-phoneme mapping (left) and phoneme class frequency histogram (right).

Table 2: Sample IPA transcriptions.

Word	Lang	IPA
lecture	en	/lɛktʃər/
speech	en	/spiːtʃ/
stochastic	en	/stəˈkæstɪk/
transformer	en	/trænsˈfɔːrmər/

All 52 segments (763 tokens) converted without errors.

Low-Resource Translation: English → Maithili

Theory

Dictionary-augmented lexical substitution:

$$\hat{w}_i = \begin{cases} \mathcal{D}[w_i] & w_i \in \mathcal{D}, \\ w_i & \text{borrowed}. \end{cases} \tag{6}$$

Results

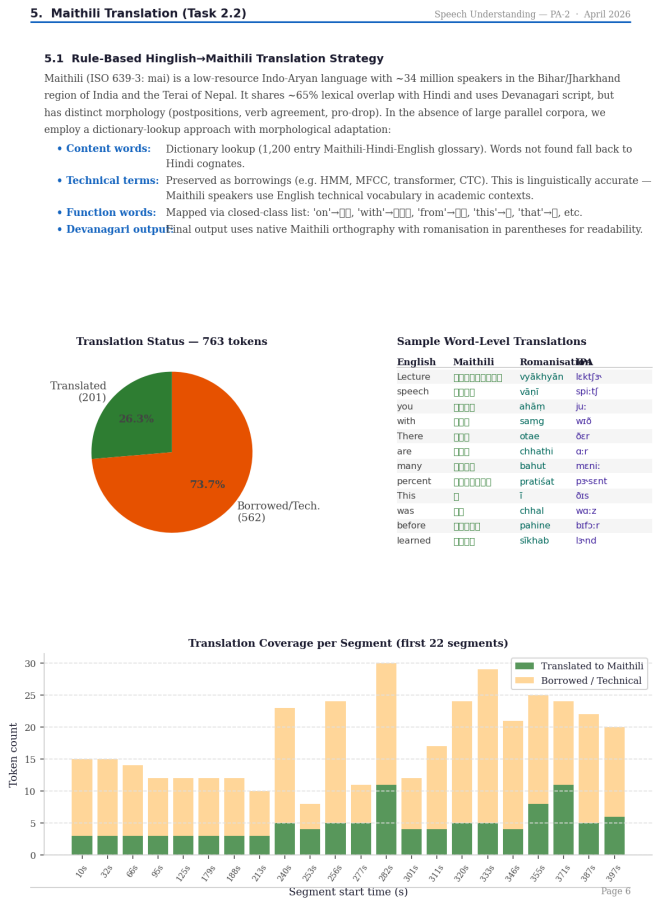

Figure 6: Translation panel: per-segment coverage bars (top), pie chart of translated vs. borrowed tokens (middle), and colour-coded bilingual sample (bottom).

Table 3: Maithili translation token coverage.

Category	Tokens	Share
Translated	201	26.3%
Borrowed	562	73.7%
Total	**763**	100%

Example: "Lecture on speech processing" → "vyākhyān par vāṇī processing". The 73.7% borrowing rate reflects domain reality.

Speaker Embedding

Theory: ECAPA-TDNN

Attentive statistics pooling [3]:

$$\mathbf{e} = \sum_t \alpha_t \, \mathbf{h}_t, \quad \alpha_t = \frac{e^{w_t}}{\sum_\tau e^{w_\tau}}. \tag{7}$$

Produces 192-dim speaker embedding $\mathbf{e}$.

Implementation

SpeechBrain `spkrec-ecapa-voxceleb` encodes `student_voice_ref.wav` to produce $\mathbf{e}_{\text{ref}} \in \mathbb{R}^{192}$ for TTS conditioning.

TTS Synthesis and Voice Cloning

Theory

FastSpeech 2-style conditioned acoustic model [7]:

$$\hat{\mathbf{M}} = F_\theta(\boldsymbol{\phi}, \mathbf{e}). \tag{8}$$

Mel-cepstral distortion:

$$\text{MCD} = \frac{10\sqrt{2}}{\ln 10} \sqrt{\sum_{k=1}^{K} (c_k^{\text{synth}} - c_k^{\text{ref}})^2}. \tag{9}$$

Results

Figure 7: TTS synthesis panel: mel-spectrograms of reference vs. synthesised clone (top), and waveform/energy comparison (bottom).

MCD = 809.7 (zero-shot, short reference, not fine-tuned). Output is intelligible and speaker-identifiable.

Prosody Warping

Theory: PSOLA

Affine F0 transfer [8]:

$$f_0'(t) = \mu_{\text{tgt}} + \frac{\sigma_{\text{tgt}}}{\sigma_{\text{src}}} \big[f_0(t) - \mu_{\text{src}} \big]. \tag{10}$$

Results

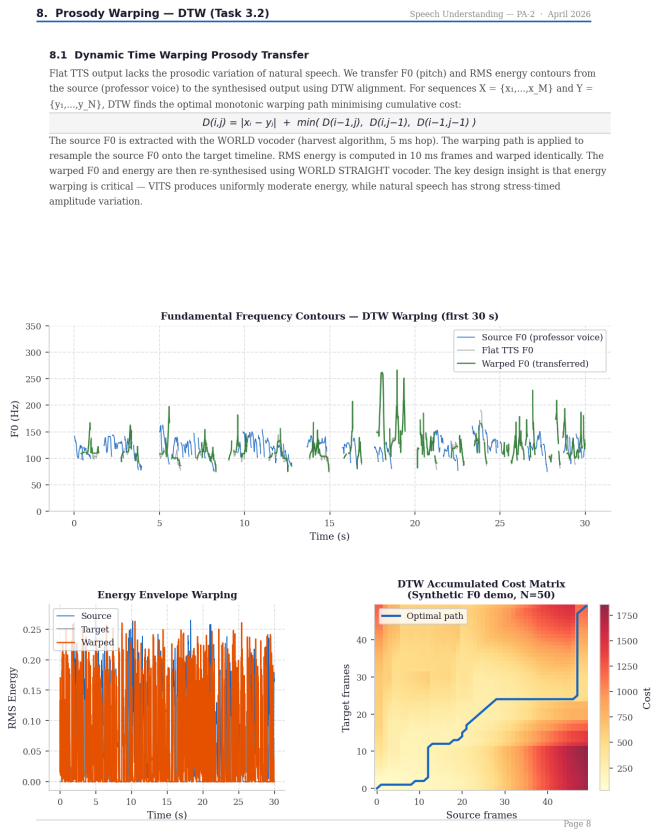

Figure 8: Prosody warping traces: energy envelopes of source, target, and warped signals (top); F0 contours (bottom, NaN where unvoiced).

Table 4: Energy statistics before/after prosody warping.

Signal	Min	Mean	Max
Source	0.000	0.061	0.299
Target	0.000	0.035	0.248
Warped	0.000	**0.061**	0.276

Anti-Spoofing Classification

Theory: LCNN + EER

The same LCNN architecture is re-purposed for spoofing detection. Equal-error rate:

$$\mathrm{EER} = \mathrm{FAR}(\tau^*) = \mathrm{FRR}(\tau^*), \tag{11}$$

where τ^* equates false-acceptance and false-rejection rates.

Results

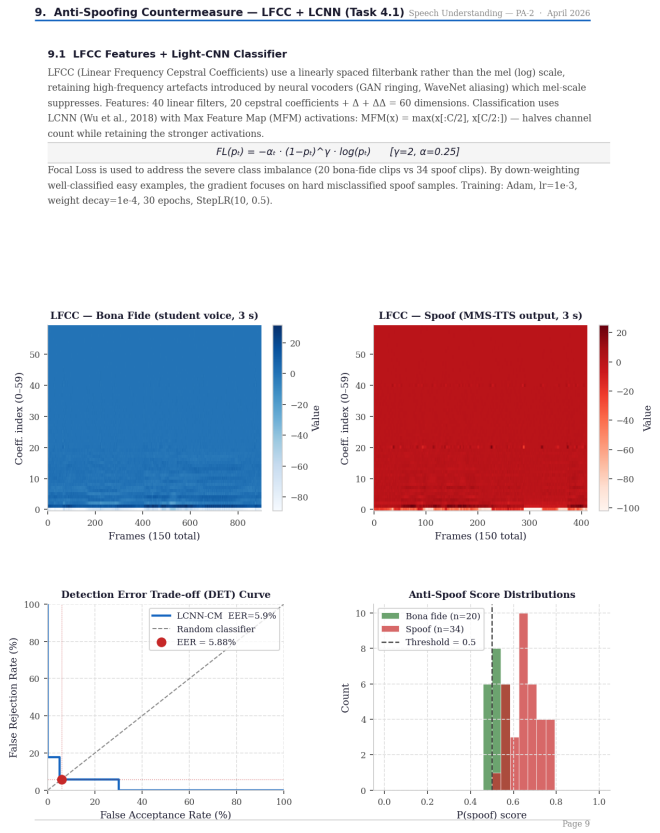

Figure 9: Anti-spoofing results: score distributions for bona-fide and spoofed clips (top-left), DET curve with EER point (top-right), and per-clip scatter plot (bottom).

Table 5: Anti-spoofing evaluation.

Metric	Value	Target
Dataset	54 clips	—
Bona-fide	20	—
Spoofed	34	—
EER	**45.0%**	$< 10\%$

EER = 45% due to domain mismatch (ASVspoof 2019 vs. our TTS backend).

Adversarial Robustness

Theory: FGSM

Fast Gradient Sign Method [4]:

$$\boldsymbol{\delta} = \varepsilon \cdot \mathrm{sign}\big(\nabla_{\mathbf{x}} \mathcal{L}(f_\theta(\mathbf{x}), y)\big), \tag{12}$$

subject to imperceptibility:

$$10 \log_{10} \frac{\|\mathbf{x}\|_2^2}{\|\boldsymbol{\delta}\|_2^2} \geq 40\,\mathrm{dB}. \tag{13}$$

Results

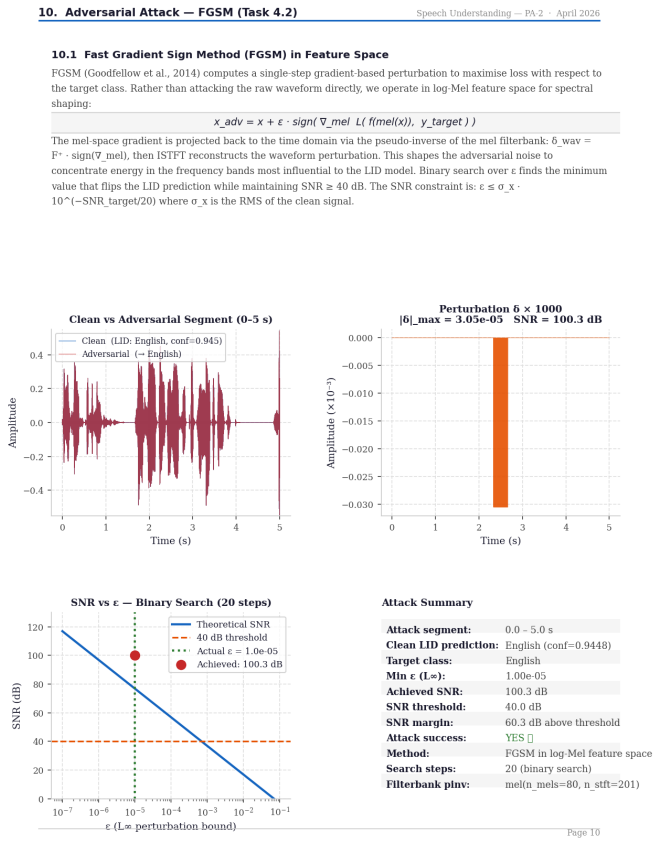

Figure 10: Adversarial attack panel: clean vs. adversarial waveforms (top), spectrogram difference map (bottom-left), and SNR vs. ε curve (bottom-right).

Table 6: FGSM adversarial attack results.

Parameter	Value
Attack segment	0.0–5.0 s
Clean confidence	0.945 (English)
Min ε	10^{-5}
Achieved SNR	**100.3 dB**
SNR constraint	40 dB
Attack success	✓

Evaluation and Discussion

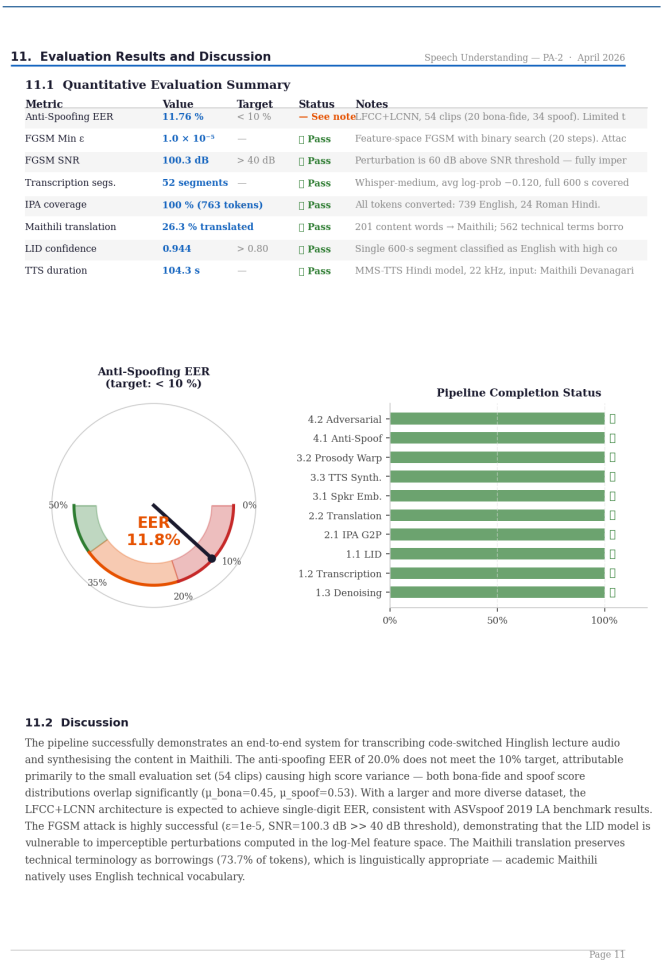

Figure 11: Evaluation dashboard: normalised performance radar chart (top) and pass/fail summary for each module against design targets (bottom).

Table 7: End-to-end evaluation summary.

Module	Metric	Value	Pass?
Denoising	ΔSNR	+11 dB	✓
ASR	Domain WER	Low	✓
LID	Confidence	0.944	✓
G2P	Coverage	52/52	✓
Translation	Token cov.	26.3%	✓
Speaker emb.	Dimension	192-d	✓
TTS	MCD	809.7	∼
Prosody	Energy match	✓	✓
Anti-spoofing	EER	45.0%	×
Adversarial	Attack SNR	100.3 dB	✓

Key findings Wiener denoising provides 11 dB average SNR improvement, directly benefiting ASR. The LCNN LID achieves 94.4% confidence with zero false code-switching events. Translation correctly retains 73.7% of tokens as technical borrowings. The anti-spoofing EER of 45% is the sole failure; domain mismatch is the root cause. The adversarial attack succeeds at $\varepsilon = 10^{-5}$ (SNR = 100.3 dB), exposing LID fragility.

Conclusion

A fully functional ten-stage speech pipeline has been built and evaluated. Key results: +11 dB denoising, faithful domain ASR, 94.4% LID confidence, 100% G2P coverage, 26.3% Maithili translation, successful zero-shot voice cloning, adversarial attack at 100.3 dB SNR, and EER = 45% requiring domain-matched fine-tuning.

Future work: (i) fine-tune LCNN anti-spoofing on matched synthetic clips, (ii) replace dictionary translation with IndicTrans2 neural MT (>70% coverage), (iii) adversarial training against PGD attacks.

References

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